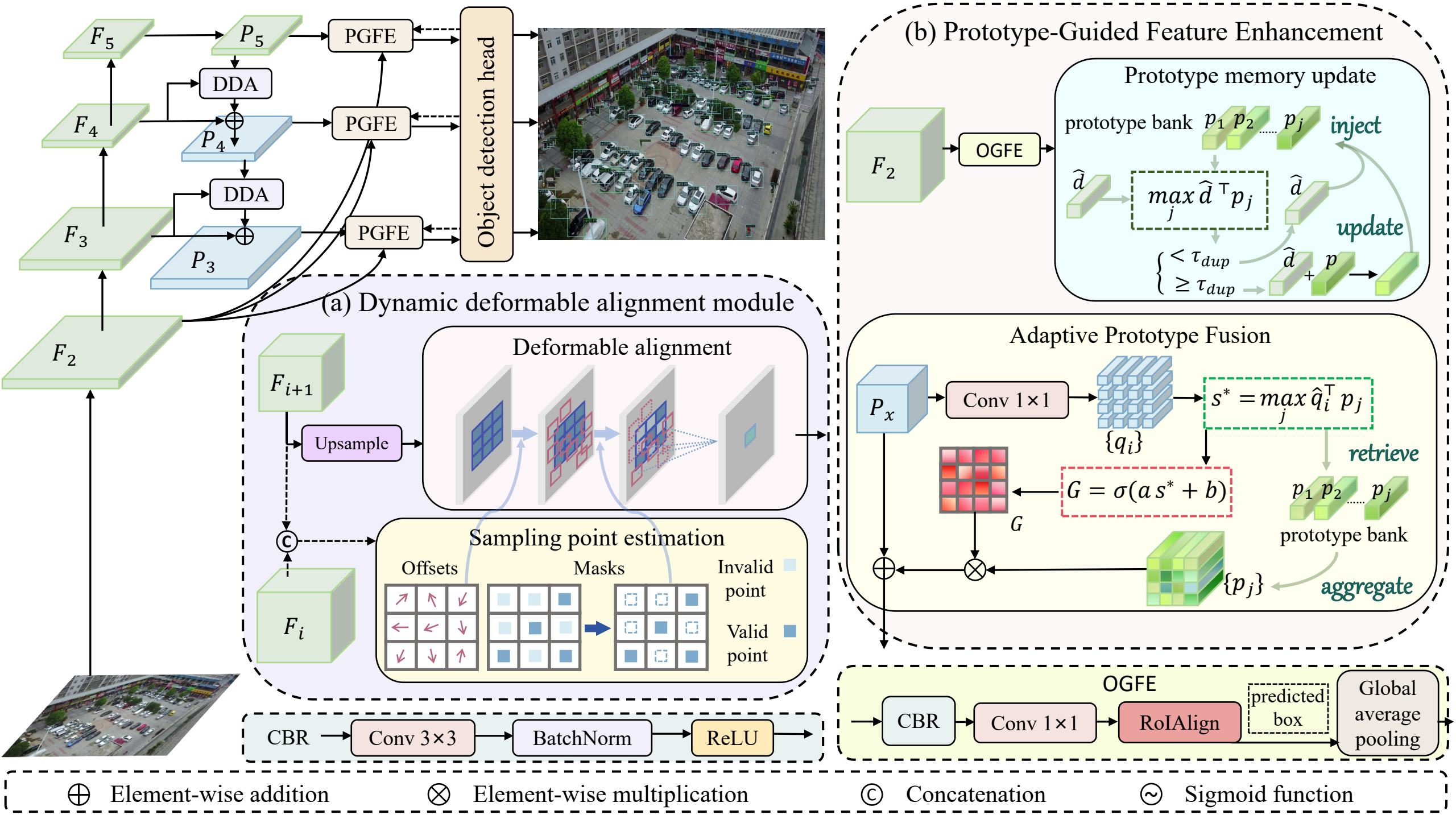

F_5
P_5
PGFE
DDA
F_4
P_4
PGFE
DDA
F_3
P_3
PGFE
F_2
Object detection head
(a) Dynamic deformable alignment module
F_{i+1}
Upsample
Deformable alignment
C
F_i
Sampling point estimation
Offsets
Masks
Invalid point
Valid point
CBR
Conv 3×3
BatchNorm
ReLU
(b) Prototype-Guided Feature Enhancement
F_2
OGFE
Prototype memory update
prototype bank p_1 p_2 ..... p_j
inject
$\hat{d}$
$\max_j \hat{d}^\top p_j$
update
$< \tau_{dup}$
$\geq \tau_{dup}$
Adaptive Prototype Fusion
P_x
Conv 1×1
$\{q_i\}$
$s^* = \max_j \hat{q}_i^\top p_j$
retrieve
$G = \sigma(a\, s^* + b)$
G
prototype bank
$\{p_j\}$
aggregate
OGFE
CBR
Conv 1×1
RoIAlign
predicted box
Global average pooling
⊕ Element-wise addition
⊗ Element-wise multiplication
© Concatenation
~ Sigmoid function